

Practical no. 14

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Class :- CO3K – B

Subject :- DSU

Program :-

```
#include<stdio.h>

#include<stdlib.h>

#include<conio.h>

struct Node{

int coefficient;

int exponent;

struct Node*next;

};

struct Node*createNode(int coeff,int exp){

struct Node*newNode=(struct Node*)malloc(sizeof(struct Node));

newNode->coefficient=coeff;

newNode->exponent=exp;

newNode->next=NULL;

return newNode;

}

struct Node*insertTerm(struct Node*head,int coeff,int exp){

struct Node*current=head;

struct Node*newNode=createNode(coeff,exp);

if(head==NULL || exp>head->exponent){

newNode->next=head;

return newNode;

}

while(current->next!=NULL&&exp<current->next->exponent){
```

```

current=current->next;

}

newNode->next=current->next;

current->next=newNode;

return head;

}

void display(struct Node*head){

struct Node*current=head;

if(head==NULL){

printf("\nList is empty!!");

}

while(current!=NULL){

printf("%dx^%d",current->coefficient,current->exponent);

current=current->next;

if(current!=NULL){

printf("+");

}

}

printf("\n");

}

void main(){

struct Node*poly1=NULL;

struct Node*poly2=NULL;

struct Node*head;

poly1=insertTerm(poly1,5,3);

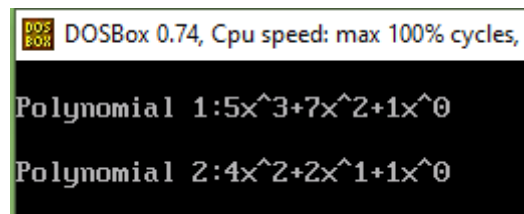
poly1=insertTerm(poly1,7,2);

poly1=insertTerm(poly1,1,0);

```

```
poly2=insertTerm(poly2,4,2);  
poly2=insertTerm(poly2,2,1);  
poly2=insertTerm(poly2,1,0);  
clrscr();  
printf("\nPolynomial 1:");  
display(poly1);  
printf("\nPolynomial 2:");  
display(poly2);  
getch();  
}
```

OUTPUT :-



DOSBox 0.74, Cpu speed: max 100% cycles,
Polynomial 1: $5x^3 + 7x^2 + 1x^0$
Polynomial 2: $4x^2 + 2x^1 + 1x^0$